Measures of Central Tendency

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Where this fits: Pipeline map step 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Descriptive statistics.
- **Compare with:** Expected value and variance of a random variable.

1. Overview

Key point: A measure of central tendency gives one number for the centre of a feature; the **mean** (G-1203), median, mode, weighted mean and trimmed mean each suit different data.

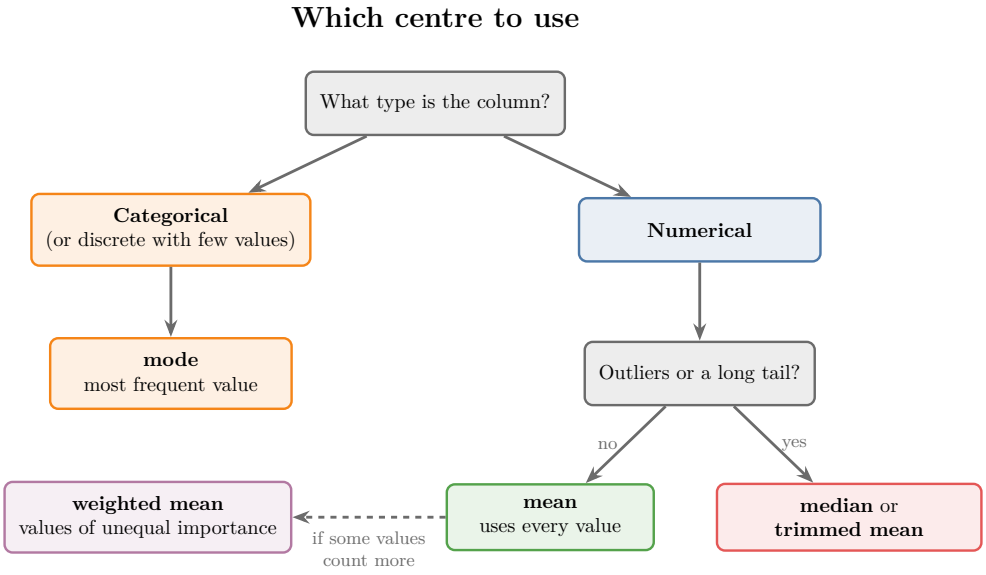


Figure 1: Choosing a measure of central tendency

Figure 1 is a flow chart: start at the top box, answer each question, and follow the arrow to the measure that fits. A **categorical feature** (G-772) has values that are labels (such as a state's name), and a **numerical** feature has values that are numbers (types of data). An **outlier** is a value far from the rest (what an outlier is). This Note works through each measure, its formula and its weak spot, in that order.

2. What central tendency means

Key point: One typical value that summarises where a feature's values sit.

A **measure of central tendency** (G-1205) is a statistical measure that represents a typical, central value of a dataset. A **feature** (G-772) is one variable of the data, one column of the

table, and an **observation** (G-1374) is one record, one row. A measure of central tendency summarises a whole feature with the single value that best represents it.

Take the `Age` feature of 1,000 passengers. Reading 1,000 ages tells us little. One number for where the ages are centred tells us a lot: "the average salary", "the average package", "a batsman's average" are all such numbers.

There are several such measures. The main ones are the **mean** (G-1203), median and mode; the weighted mean and trimmed mean are useful variants.

3. Mean

> **Key point:** The mean is the sum of the values divided by how many there are; it uses every value, so one extreme value can drag it far away.

Five students score 3, 4, 1, 2 and 5 marks in a quiz. Share the marks out so that everyone ends up with the same score: this equal share is the **mean** (G-1203). We find it in two steps, and count, mean, standard deviation, minimum and maximum in code covers it too.

Figure 2 draws the five scores as equal **weights** (G-2111) on a beam. The orange triangle marks the one place where the beam balances.

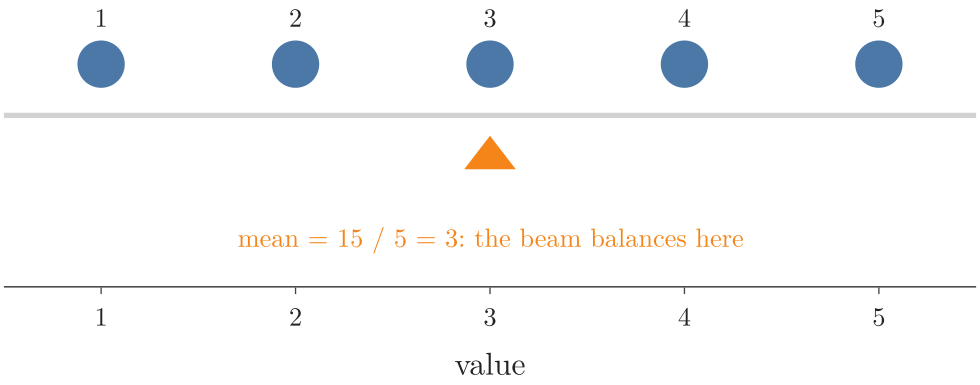


Figure 2: The scores 3, 4, 1, 2, 5 as equal weights on a beam: the mean, 3, is the balance point

The two steps, one per line:

Step	Work	Result
1. Add all the values	$3+4+1+2+5$	15
2. Divide by how many values there are	$15/5$	3

Now the formula, with every symbol named. Call the first score $x_1 = 3$, the second $x_2 = 4$, and so on up to $x_5 = 5$. The little number below x is only the position of the score in the list, and the number of scores is $n = 5$. The sign $\sum$ ("sigma") means "add up", so $\sum_{i=1}^{5} x_i$ is $x_1 + x_2 + x_3 + x_4 + x_5$.

$$\sum_{i=1}^{5} x_i = 3 + 4 + 1 + 2 + 5 = 15$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i = \frac{1}{5} \times 15 = 3$$

The result matches the balance point in Figure 2. The bar over x is read "x bar" and stands for the sample mean.

The same arithmetic has two names, depending on what the values are. Parameters and statistics explains why. A **population** is every value we care about, say all N students of a college; a **sample** is the n values we actually measured.

	Values	Count	Symbol	Formula
Population mean (G-1524)	the whole population	N	μ ("mu")	$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$
Sample mean (G-1725)	one sample	n	$\bar{x}$	$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

μ is the true centre of the whole population; $\bar{x}$ is the centre of one sample. Nothing guarantees they are equal: they can be close, or very different.

3.1 The weak spot of the mean: outliers

> **Key point:** One very high or very low value pulls the **mean** (G-1203) towards it, so the mean stops describing anyone.

As one outlier changes the mean shows, one billionaire in a classroom drags the mean salary into the crores. Here is the same effect on a class we will reuse below.

Nine students earn between 28 and 40 thousand rupees a month. A tenth classmate skipped placements, founded a start-up and now earns 20 lakh rupees (2,000 thousand) a month. Figure 3 shows what happens to the mean.

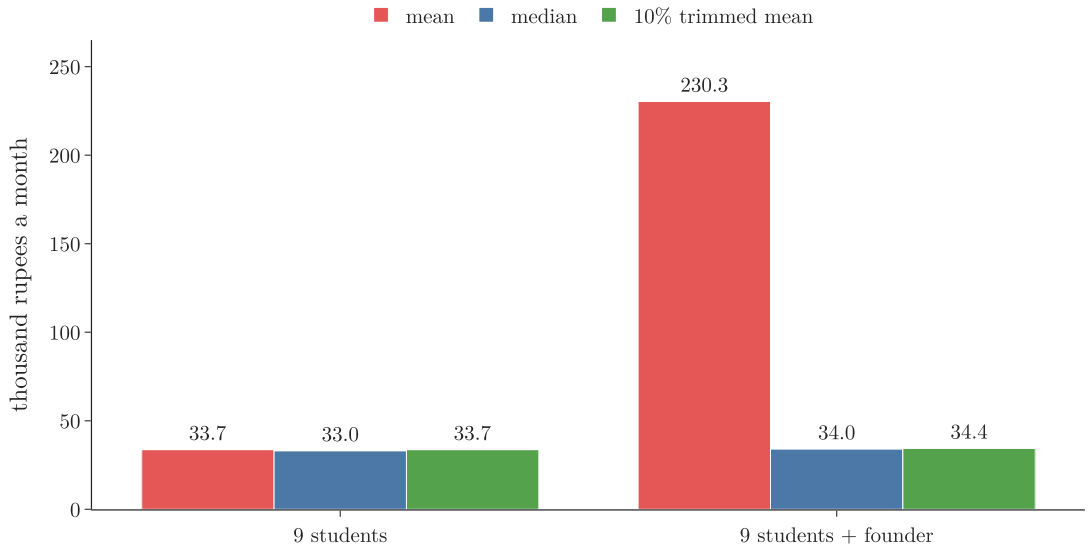


Figure 3: Mean, median and 10% trimmed mean before and after one very high earner joins

Class	Sum of salaries	Divide by the count	Mean
Nine students	28+30+31+32+33+ 35+36+38+40 = 303	303/9	33.7 thousand rupees, a fair summary
With the founder	303 + 2000 = 2303	2303/10	230.3 thousand rupees: nobody in the class earns anything like that

So before using the mean, we check whether the **feature** (G-772) has outliers. If it does, the mean is not a good summary of that feature.

4. Median

> **Key point:** The median is the middle value of the sorted data; extreme values sit at the ends of the sorted list, so they cannot move it.

Line the students up in order of marks and pick the one standing in the middle. That student's mark is the **median** (G-1209), the middle value of the sorted data. Percentiles in code covers it too.

Figure 4 shows two lists. The top row has 5 values, so one value sits in the middle. The bottom row has 6 values, so two values share the middle, and we take their **mean** (G-1203).

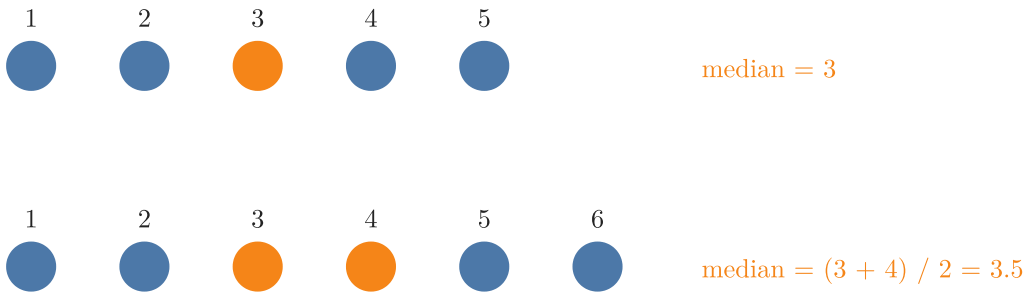


Figure 4: Sorted values 1 to 5 and 1 to 6: the middle value, or the mean of the two middle values, is the median

Worked steps for the top row, 1, 2, 3, 4, 5 (5 values):

Step	Work	Result
1. Sort the values	already in order	1, 2, 3, 4, 5
2. Count them	five values	$n = 5$
3. Find the middle position	$(5 + 1)/2$	position 3
4. Read the value there	third value	3

Add a sixth value, 6. Now there is no single middle value:

Step	Work	Result
1. Sort the values	already in order	1, 2, 3, 4, 5, 6
2. Count them	six values	$n = 6$
3. Find the two middle positions	$6/2$ and $6/2 + 1$	positions 3 and 4
4. Read the values there	third and fourth	3 and 4
5. Take their mean	$(3 + 4)/2$	3.5

The formal version names the sorted values $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$. The bracket in $x_{(3)}$ means "the third value after sorting"; here $x_{(3)} = 3$. There are two cases:

For odd n:

$$\text{median} = x_{((n+1)/2)}$$

For even n:

$$\text{median} = \frac{x_{(n/2)} + x_{(n/2+1)}}{2}$$

For $n = 5$ (odd), the middle value is $x_{(3)} = 3$. For $n = 6$ (even), average the two middle values:

$$\frac{x_{(3)} + x_{(4)}}{2} = \frac{3 + 4}{2} = 3.5$$

Both reproduce the table rows.

Now replace the 6 by 60,000. The sorted list is 1, 2, 3, 4, 5, 60000, the middle positions are still 3 and 4, and

$$\text{median} = \frac{3 + 4}{2} = 3.5$$

The unchanged median of 3.5 shows why the median resists outliers. However large an extreme value is, sorting puts it at the end of the list, and the middle stays where it was. In Figure 3, the founder moves the median only from 33 to 34 thousand rupees.

Figure 5 drags the tenth salary up step by step, from 40 to 2,000 thousand rupees. Watch the red mean climb with it while the blue median and the green **trimmed mean** (G-2017) (section 7) stay flat at about 34. Both horizontal axes use a **log scale**: the step from 30 to 300 takes the same width as the step from 100 to 1000, because each is ten times bigger. That lets 40 and 2,000 fit on one axis.

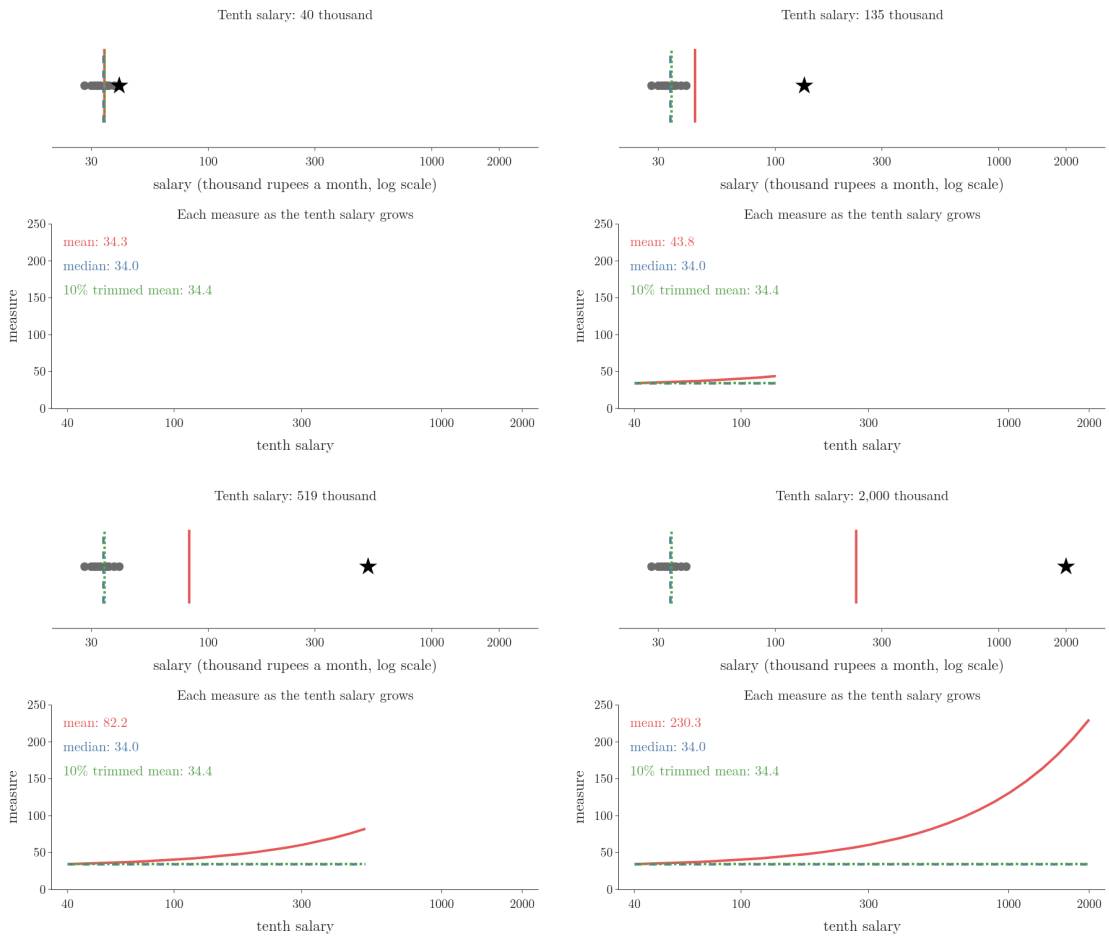


Figure 5: The tenth salary dragged from 40 to 2,000 thousand rupees: the mean follows it up to 230.3, the median stays at 34 and the 10% trimmed mean at 34.4

The same reasoning gives practical advice: when comparing colleges or companies, look at the median package, not the average one. One student with a package of 1 crore raises the average for everyone, but leaves the median almost untouched.

Python: Mean and median.

```
import numpy as np

x = np.array([1, 2, 3, 4, 5, 60000])
np.mean(x)      # 10002.5
np.median(x)    # 3.5
```

On a pandas column we write `df["Age"].mean()` and `df["Age"].median()`.

5. Mode

> **Key point:** The mode is the most frequent value; it is the natural centre for categorical and discrete data.

Ask a class "which state are you from?" and count the hands. The state with the most hands is the **mode** (G-1251), the value that appears most often in the data. Counting is the only work, so the mode also suits data that are not numbers.

Take the values 1, 2, 1, 3, 1, 4, 2, 1. Figure 6 draws one bar per distinct value, as tall as its count. The tallest bar, in orange, is the mode.

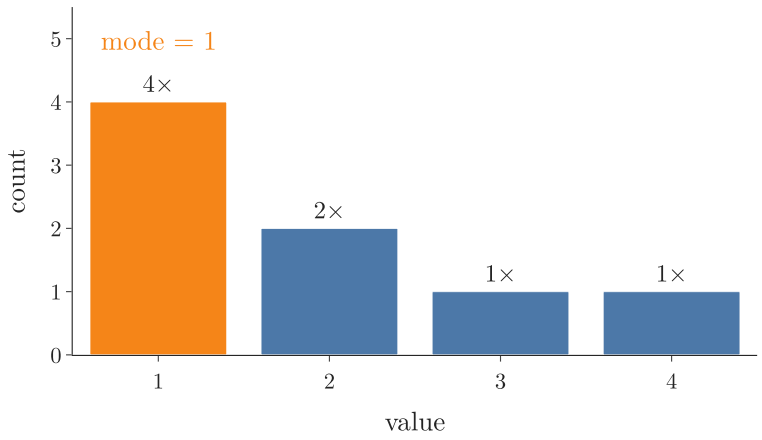


Figure 6: Counting each value of 1, 2, 1, 3, 1, 4, 2, 1: the tallest bar is the mode

Worked steps, one per row:

Value	Count (tally)	How many times
1	four times	4
2	twice	2
3	once	1
4	once	1

The largest count is 4, so the mode is 1.

The formal version needs one new symbol. Let $f(v)$ be the number of times the value v appears: here $f(1) = 4$, $f(2) = 2$, $f(3) = 1$ and $f(4) = 1$.

$$\text{mode} = \text{the value } v \text{ with the largest } f(v)$$

Here $f(1) = 4$ is the largest, so the mode is $v = 1$, as in the table.

The mode is most useful for:

- **Categorical features** (labels, not numbers), where the **mean** (G-1203) and median make no sense. Ask a class which state each student comes from and count; if Maharashtra comes up most often, it is the mode.
- **Discrete features with few values,** such as the number of siblings on board.

For a continuous **feature** (G-772), the mode is rarely useful: with values such as 32.17 and 32.18, almost every value appears only once.

If two values tie for the highest count, both are modes. Data with two modes is **bimodal** (G-296) (two peaks, as in how normal each feature is); with more, it is **multimodal** (G-1275).

> **Python:** The mode.
> ```
> import pandas as pd
>
> pd.Series([1, 2, 1, 3, 1, 4, 2, 1]).mode()    # 1
> pd.Series([1, 1, 2, 2, 3]).mode()             # 1 and 2: a tie
> ```
> `mode()` returns every value that ties for first place, so its result is a Series, not a single number.

6. Weighted mean

> **Key point:** The weighted mean multiplies each value by its importance before averaging, so more important values count more.

In the ordinary **mean** (G-1203), every value counts equally. Sometimes one value deserves more say than another. We predict a house price with three models: linear regression says 10 lakh rupees, a random forest 15 lakh, and XGBoost 12 lakh. From past results, XGBoost has been right most often, so we trust it most. We give each prediction a **weight** (G-2111), a number that says how much it counts, and average with those weights: the **weighted mean** (G-2117).

Figure 7 shows the idea. Each prediction is a dot on a beam, and a heavier dot is a bigger weight. The weighted mean (orange) is the balance point, pulled towards the heaviest dot.

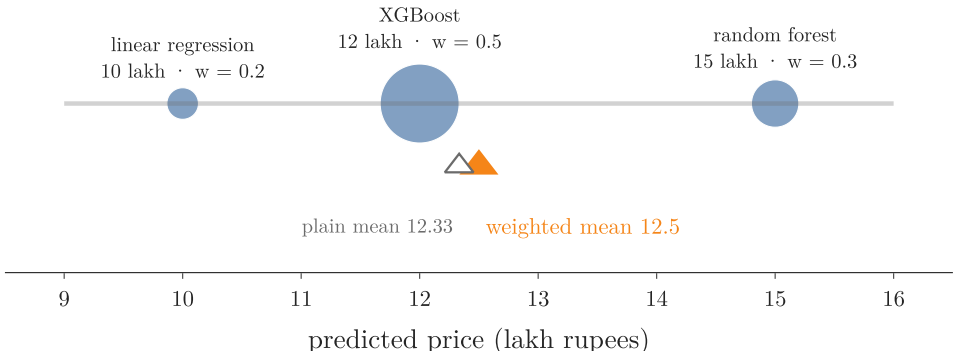


Figure 7: The three predictions as weights on a beam, each dot sized by its weight: the weighted mean (orange) is the balance point, pulled towards the heaviest dot

The weights are 0.2 for linear regression, 0.3 for the random forest and 0.5 for XGBoost. Worked steps, one per row:

Step	Work	Result
1. Multiply each prediction by its weight	0.2×10	2
	0.3×15	4.5
	0.5×12	6
2. Add these products	$2 + 4.5 + 6$	12.5
3. Add the weights	$0.2 + 0.3 + 0.5$	1
4. Divide the sum of products by the sum of weights	$12.5/1$	12.5 lakh rupees

For comparison, the plain mean of the three values is:

$$\frac{10 + 15 + 12}{3} = 12.33 \text{ lakh}$$

The weighted mean leans towards XGBoost.

The formal version names the values $x_1 = 10$, $x_2 = 15$, $x_3 = 12$ and their weights $w_1 = 0.2$, $w_2 = 0.3$, $w_3 = 0.5$:

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i \, x_i}{\sum_{i=1}^{n} w_i}$$

The top is step 2 (12.5), the bottom is step 3 (1), so $\bar{x}_w = 12.5$, as in the table.

This weighting is exactly what a voting regressor with weights does (see weights in a voting regressor). The ordinary mean is the special case where every weight is equal.

> **Python:** The weighted mean.
> `np.average([10, 15, 12], weights=[0.2, 0.3, 0.5])  # 12.5`

7. Trimmed mean

> **Key point:** The trimmed mean drops a fixed share of the smallest and largest values, then averages the rest, so outliers cannot pull it.

In a diving contest, seven judges score a dive. One judge may be far too generous, or far too harsh. To stop that one judge deciding the result, we throw away the highest scores and the lowest scores and average the rest. This is the **trimmed mean** (G-2017): remove a chosen percentage of the smallest and of the largest values, then take the mean of what is left. The percentage removed from each end is the **trimming percentage** (G-2018).

We use the class from section 3.1: nine students and one founder. Sorted, the $n = 10$ salaries in thousand rupees are 28, 30, 31, 32, 33, 35, 36, 38, 40 and 2000. Figure 8 shows the trim on a log scale (equal distances along the horizontal axis **mean** (G-1203) equal factors, as the ticks 25, 30, 40, 50, 100, 200, 500, 1000, 2000 show, so 28 and 2000 fit on one axis): the lowest and highest salaries are cut, and the mean is taken of the 8 left.

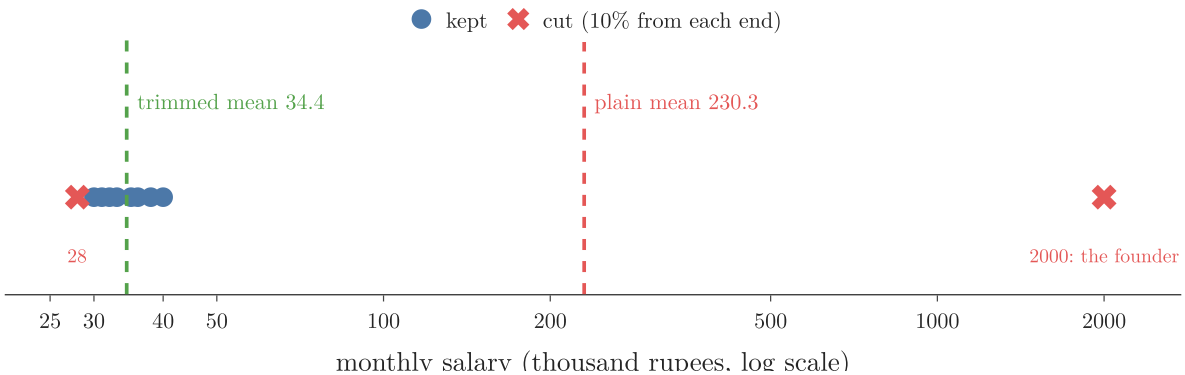


Figure 8: The worked example on a log scale: the 10% trim cuts the lowest salary, 28, and the founder's 2000; the mean of the 8 left is 34.4, while the plain mean of all 10 is 230.3, above every student but one

Worked steps for a 10% trim, one per row:

Step	Work	Result
1. Sort the values	already sorted	28, 30, . . . , 40, 2000
2. Work out how many to cut from each end	0.1×10	1 value

Step	Work	Result
3. Cut the lowest and the highest	remove 28 and 2000	8 values left
4. Add the values left	$30 + 31 + 32 + 33 + 35 + 36 + 38 + 40$	275
5. Divide by how many are left	275/8	34.375

The plain mean was 230.3; the trimmed mean, 34.4, describes the class again (Figure 3, and Figure 8).

The formal version uses the sorted values $x_{(1)} \leq \cdots \leq x_{(n)}$ and the trimming percentage p written as a fraction ($p = 0.1$ for 10%). The number of values cut from each end is $k = \lfloor p\, n \rfloor$, where $\lfloor\ \rfloor$ means round down. For $n = 10$ and $p = 0.1$:

$$\lfloor 0.1 \times 10 \rfloor = \lfloor 1 \rfloor = 1$$

$$\bar{x}_{\text{trim}} = \frac{1}{n - 2k} \sum_{i=k+1}^{n-k} x_{(i)}$$

With $n = 10$ and $k = 1$ the sum runs from $x_{(2)} = 30$ to $x_{(9)} = 40$, which is 275. The count left is:

$$n - 2k = 10 - 2 = 8$$

$$\bar{x}_{\text{trim}} = \frac{275}{8} = 34.375$$

This is the value in the table.

The trimmed mean sits between the mean and the **median** (G-1209): trimming nothing gives the plain mean, and trimming almost 50% from each end leaves only the middle, the median. In between, it uses more values than the median yet ignores the extremes.

How much to trim depends on the data. We look at its distribution first, for example with a box plot, see where the outliers are, and choose the trimming percentage from that.

> **Extra:** With 9 students, a 10% trim cuts $\lfloor 0.9 \rfloor = 0$ values, so in Figure 3 the trimmed mean of the nine equals their plain mean, 33.7. A common general-purpose choice is 20% from each end (Wilcox 2012). The median is the extreme case, a trim of almost 50% from each end.

Extra: Judged sports use a trimmed mean, so one very generous or very harsh judge cannot decide the result. In diving, the two highest and two lowest of seven scores are dropped (USA Diving); in gymnastics, the highest and lowest execution scores are dropped (FIG Code of Points).

Python: The trimmed mean.

```
from scipy import stats

salaries = [28, 30, 31, 32, 33, 35, 36, 38, 40, 2000]
stats.trim_mean(salaries, 0.1)   # 34.375
```

`0.1` is the share cut from each end. `trim_mean` sorts the values itself.

8. Choosing a measure

Key point: Use the mode for categories, the **mean** (G-1203) for numbers without outliers, and the median or trimmed mean when outliers are present.

Measure	Uses	Affected by outliers?	Best for
Mean	every value	yes, strongly	numerical data without outliers
Median	the middle value(s)	almost not	numerical data with outliers or skew; ordinal data (ordered labels, such as small, medium, large)
Mode	the counts	no	categorical and discrete data
Weighted mean	every value and its weight	yes	values of unequal importance
Trimmed mean	the values left after trimming	only if trimming is too light	numerical data with a few outliers

When there are no outliers, the mean is the better summary: it uses every value, while the **median** (G-1209) uses only one or two. No rule says one measure is always best; we look at the data first. For a skewed feature, skewness in univariate analysis shows how the mean is pulled towards the long tail.

Extra: Two more means appear in special cases.

- **Geometric mean** (G-844) for growth rates: the n-th root of the product of the values.

$$(x_1 x_2 \cdots x_n)^{1/n}$$

An investment grows 10% one year (factor 1.1) and 50% the next (factor 1.5). One

step per line:

$$1.1 \times 1.5 = 1.65$$

$$\sqrt{1.65} \approx 1.2845$$

The geometric mean factor, 1.2845, is an average growth of 28.45% a year. The ordinary mean, 30%, is wrong:

$$1.3 \times 1.3 = 1.69 \neq 1.65$$

- **Harmonic mean** (G-880) for rates such as speeds: n divided by the sum of the reciprocals of the values.

$$\frac{n}{1/x_1 + \cdots + 1/x_n}$$

The F1 score is the harmonic mean of precision and recall (see why the harmonic mean; precision and recall are two scores of a classifier).

9. Summary

Measure	Formula	Example (values)	Result	Why it matters
Mean	$\sum x_i/n$	3, 4, 1, 2, 5	3	uses every value, so it is the best summary when there are no outliers
Median	middle of sorted values	1, 2, 3, 4, 5, 60000	3.5	sorting puts extreme values at the ends, so they cannot move it
Mode	most frequent value	1, 2, 1, 3, 1, 4, 2, 1	1	needs only counting, so it works for categories
Weighted mean	$\sum w_i x_i / \sum w_i$	10, 15, 12 with weights 0.2, 0.3, 0.5	12.5	lets the more trusted values count more
Trimmed mean (10%)	mean after cutting each end	the 10 class salaries	34.375	cuts the extremes first, so the founder's salary cannot pull it

- **Population mean** (G-1524) μ and **sample mean** (G-1725) $\bar{x}$ use the same arithmetic but describe different things, so a sample mean need not equal the population mean.
- The **mean** (G-1203) is pulled by outliers; the median and the trimmed mean are not, so check a feature for outliers before using its mean (compare median packages, not average ones).
- Ties give several **modes** (G-1251): bimodal or multimodal data, which is why pandas `mode()` returns a Series, not a single number.
- No measure is always best, so look at the data first and follow Figure 1 to the measure that fits it.

10. Sources

Built from

- CampusX, "Session 38 - Descriptive Statistics Part 1 | DSMP 2023", YouTube, https://www.youtube.com/watch?v=Uv3Blie7F3g

Other references

- Wilcox, R. R. (2012). *Introduction to Robust Estimation and Hypothesis Testing*, 3rd ed. Academic Press. Chapter 3, the trimmed mean (20% trimming as a general-purpose choice).
- FIG (Fédération Internationale de Gymnastique). *Rhythmic Gymnastics Code of Points 2022-2024*: section 3.6.1, Execution panel: the highest and lowest of four scores are eliminated.
- USA Diving. *Judging and Scoring*. usadiving.org, Diving 101.

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Measure of central tendency (G-1205)	A single number for the typical, central value of a column.
Weight (in a weighted mean) (G-2111)	A number saying how much a value counts in a weighted mean.
Population mean (μ) (G-1524)	The mean of every value in the population.
Sample mean ($\bar{x}$) (G-1725)	The average of the values in a sample, $\bar{x}$: add them up and divide by n; it is used as the estimate of the population mean μ.
Median (G-1209)	The middle value of sorted data; the 50% percentile.
Trimmed mean (G-2017)	The mean after removing a fixed share of the smallest and largest values.
Mode (G-1251)	The value that appears most often in the data, so it sits under the highest peak of a histogram or a distribution curve; it gives a centre for data that are not numbers, and a belief with several peaks has several modes.
Multimodal (G-1275)	Having more than one mode, that is more than one peak (two peaks: bimodal), such as a robot's belief peaking at three doors; it shows the data point to several separate values, so one mean or one best guess would hide them.
Weighted mean (G-2117)	A mean in which each value is multiplied by a weight saying how much it counts.
Trimming percentage (G-2018)	The share of values removed from each end for a trimmed mean.
Geometric mean (G-844)	An average for values that multiply, such as growth factors: multiply the n values together and take the n-th root.

Term	Meaning
Mean (G-1203)	The average of the values; the centre of the data.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Bimodal (G-296)	A distribution with two peaks.
Harmonic mean (G-880)	An average that stays close to the smaller of the values, so one low value pulls it down; F1 uses it to combine precision and recall. For two values: $2ab/(a+b)$.